German Petrol Prices and Forecast Evaluation

Historical price reconstruction, intraday patterns and forecast benchmarking

IPDA Assignment 2

FH Dortmund

Analysis period: 1 May–30 September 2026

Stations: HEM, STAR, SHELL and ARAL

Research Questions and Objectives

- How do petrol prices vary by hour?
- Are there differences between weekdays and weekends?
- How do the four stations compare?
- How often do prices increase?
- How accurate are the forecasts?
- Do forecasts outperform a previous-day baseline?

Historical events → Hourly reconstruction → Pattern analysis → Forecast evaluation → Baseline comparison

To combine historical price analysis with a fair forecast evaluation.

Data Sources and Study Scope

HISTORICAL

153
daily files

10,498
price events

14,688
hourly observations

Historical period:
1 May–30 September 2026

FORECAST

514
files

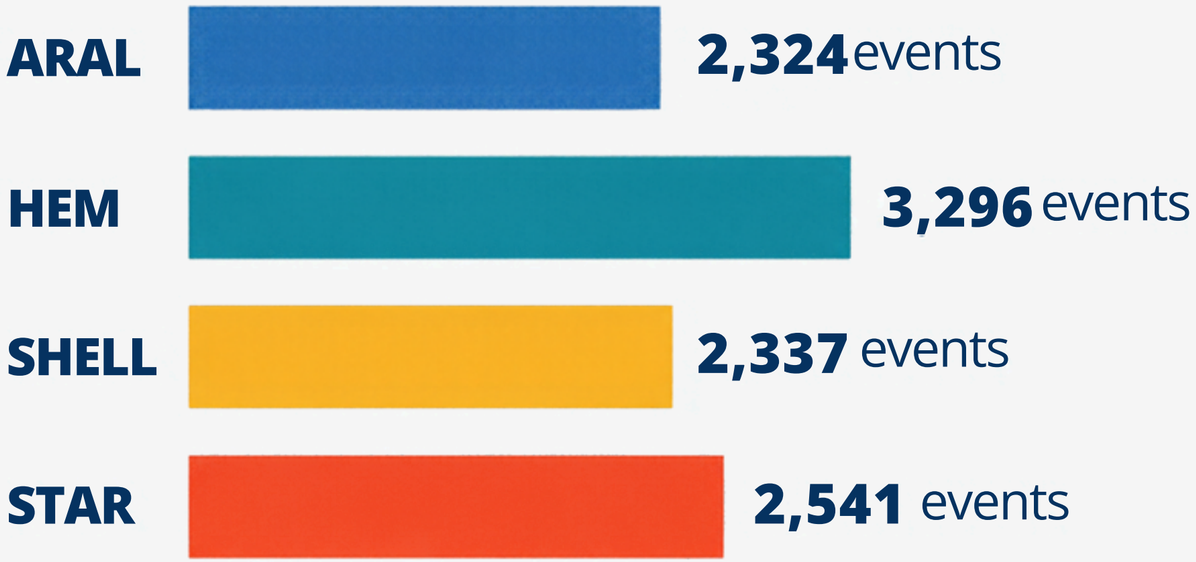
759,408
records

Forecast targets:
5 August–18 October 2026

TruckDiesel has no matching observed historical series.

Data Audit and Station Verification

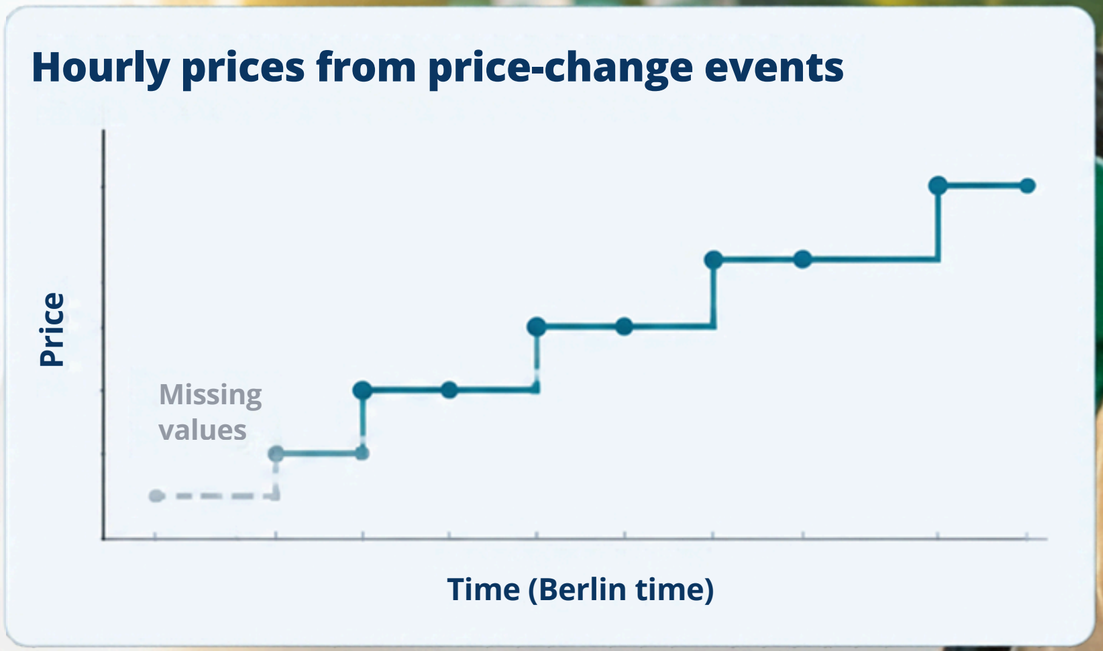
HISTORICAL EVENT COUNTS



- **No duplicate station/time events.**
- **No zero prices or suspicious prices below €1 or above €3.**
- **All four assigned station IDs were verified.**
- **Station metadata was checked against the September snapshot.**

Price Reconstruction Method

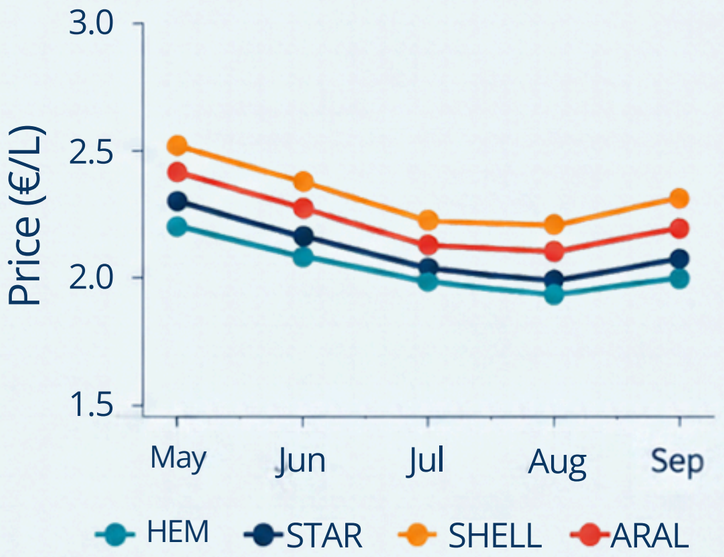
1. Convert timestampss to Berlin time.
2. Create an hourly time grid.
3. Apply the last known price forward.
4. Preserve missing values before the first event.
5. Calculate hourly, daily, weekly and monthly summaries.



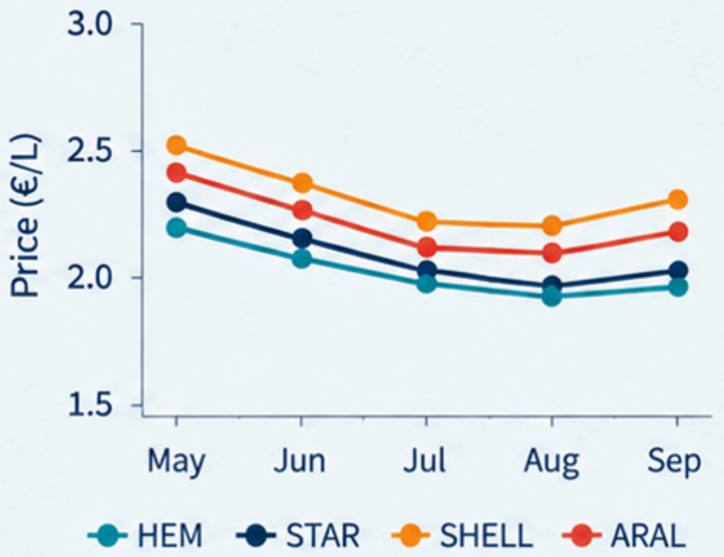
14,688 hourly station observations

Historical Price Levels and Monthly Evolution

Monthly mean E10 price by station



Monthly mean Diesel price by station



Price ranges (across all stations)

Diesel:
€1.619–€2.629/L

E10:
€1.729–€2.489/L

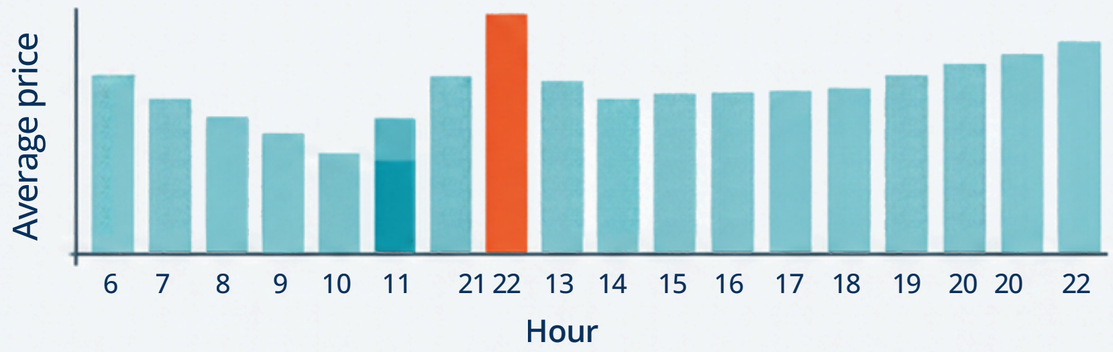
E5:
€1.789–€2.549/L

How did prices change from May to September?

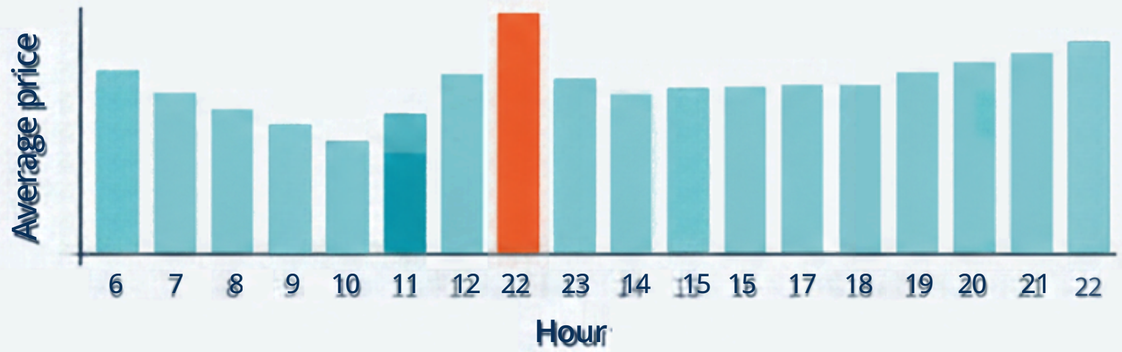
Did stations move together, and how did Diesel compare with E10?

Intraday Price Patterns

Averagay price by hour – E10



Average price by hour – Diesel



Mean daily spread by station – E10

STATION	CHEAPEST HOUR	COSTLIEST HOUR	DAILY SPREAD
ARAL	11:00	13:00	20.89 cents/L
HEM	12:00	13:00	19.23 cents/L
SHELL	12:00	13:00	19.01 cents/L
STAR	12:00	13:00	18.34 cents/L

Mean daily spread by station – Diesel

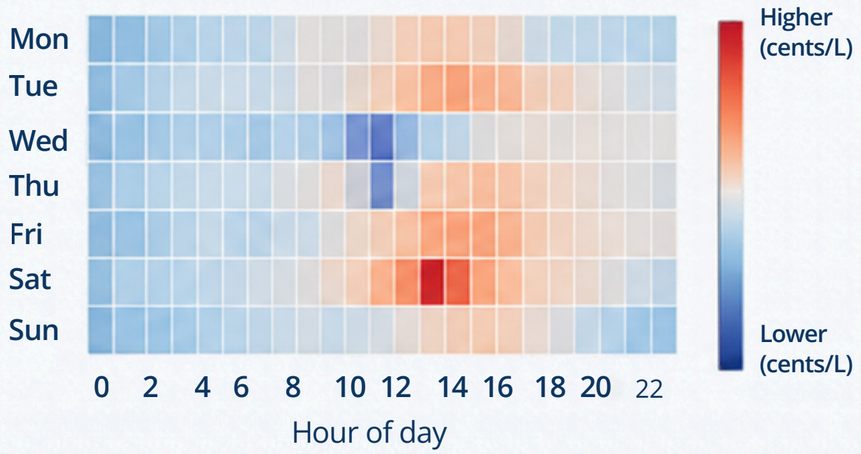
STATION	CHEAPEST HOUR	COSTLIEST HOUR	DAILY SPREAD
ARAL	11:00	13:00	22.86 cents/L
HEM	12:00	13:00	20.12 cents/L
SHELL	12:00	13:00	21.56 cents/L
STAR	12:00	13:00	20.84 cents/L

TAKEAWAY

Lowest around 11:00–12:00; highest around 13:00.
Diesel had larger spreads at every station.

Weekday/Hour Patterns and Price Increases

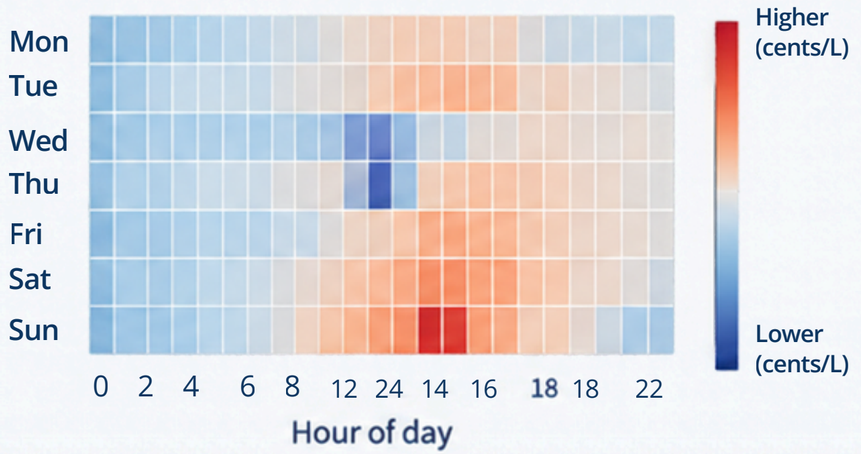
E10



E10 low:
Thursday 12:00,
–4.65 cents/L

E10 high:
Friday 13:00,
+15.37 cents/L

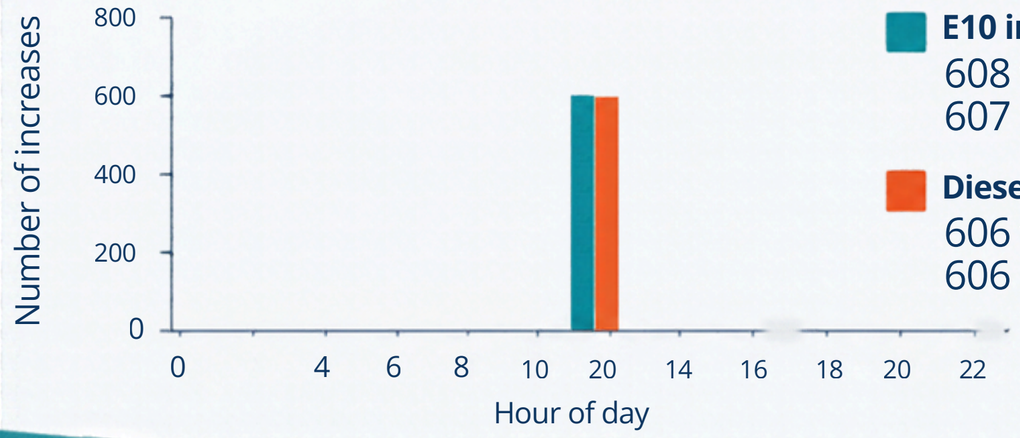
Diesel



Diesel low:
Thursday 12:00,
–5.06 cents/L

Diesel high:
Sunday 13:00,
+17.21 cents/L

Price increases by hour



E10 increases:
608 total;
607 at 12:00

Diesel increases:
606 total;
606 at 12:00

Takeaway

Price increases were overwhelmingly concentrated at 12:00; this is observed, not causal evidence.

Forecast Data and Evaluation Method

Forecast retrieval → **1-2 hour horizon** → **target time** → **actual active price**

- 759,408 forecast records across four stations.
- The latest eligible forecast was selected for each target.
- Actual prices were matched with a backward as-of join.
- 222 evaluation points per station/fuel combination.
- Only SuperE10 and Diesel were evaluated.

METRIC DEFINITIONS

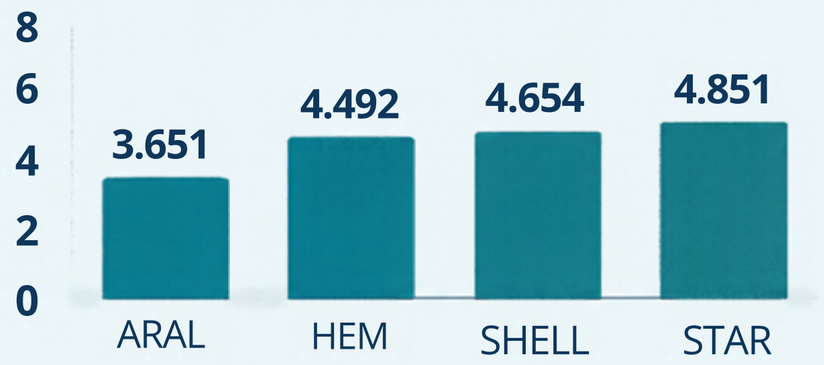
MAE: average absolute error

RMSE: penalises large errors

Bias: overprediction or underprediction

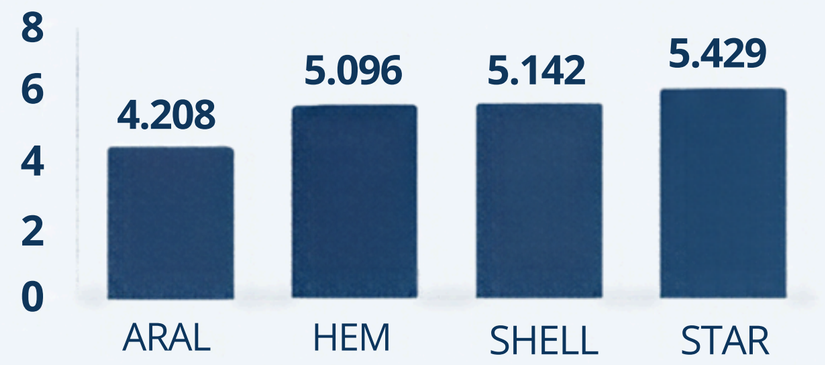
Forecast Accuracy by Station

SuperE10



ARAL	MAE 3.651, RMSE 4.873, bias +0.152
HEM	MAE 4.492, RMSE 6.521, bias +0.764
SHELL	MAE 4.654, RMSE 7.089, bias +0.709
STAR	MAE 4.851, RMSE 7.183, bias +0.284

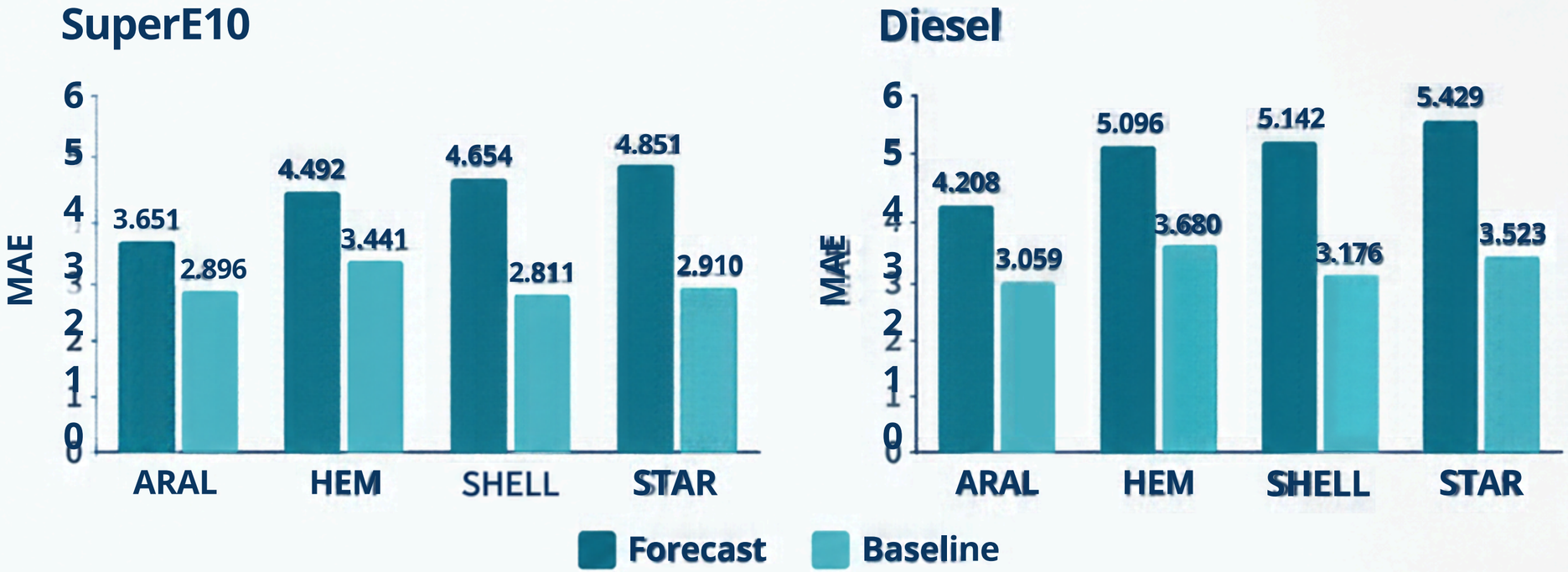
Diesel



ARAL	MAE 4.208, RMSE 5.425, bias −0.072
HEM	MAE 5.096, RMSE 6.861, bias +0.757
SHELL	MAE 5.142, RMSE 7.540, bias +0.611
STAR	MAE 5.429, RMSE 7.941, bias +0.235

ARAL had the lowest error; STAR the highest.
SuperE10 forecasts were more accurate than Diesel forecasts.

Forecast Versus Baseline



Forecast Baseline

Baseline

Tomorrow's price at a given time equals the price observed one day earlier at the same time.

The previous-day same-time baseline outperformed the supplied forecast at all four stations for both fuels.

Macro-average across four stations

Model	Fuel	MAE	RMSE	Bias
Forecast	SuperE10	4.412	6.417	+0.477
Baseline	SuperE10	3.015		
Forecast	Diesel	4.969	6.942	+0.383
Baseline	Diesel	3.360		

Conclusions

Lowest prices around 11:00–12:00.
Highest around 13:00.
Increases concentrated at 12:00.
Diesel had larger spreads.
The baseline outperformed for both fuels.

Limitations

Reconstructed event data.
Unknown pre-event price.
Shorter forecast coverage.
Hourly targets do not represent every minute.
No observed TruckDiesel series.
Baseline RMSE and bias not calculated.
Descriptive, not causal.

A simple previous-day same-time baseline was more accurate than the supplied short-horizon forecast in this evaluation.